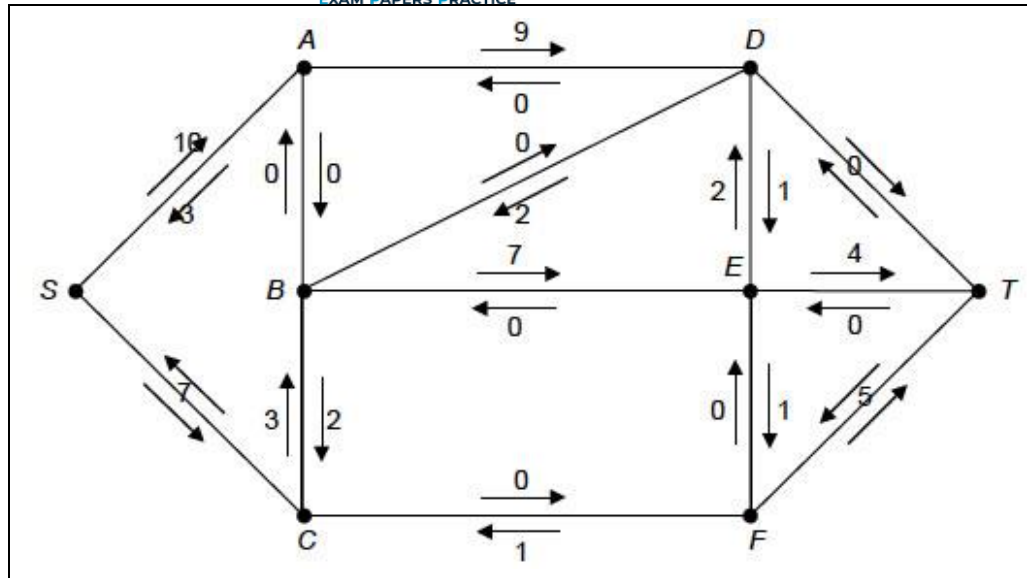


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Explains correctly using maximum flow into and minimum flow out of A	AO2.4	E1	Max flow into vertex $A = 7$ Min flow out of vertex $A = 4 + 3 = 7$ As flow into $A =$ flow out of A , both AD and AB must be at the lower capacity
(i)				
(ii)	Explains correctly using maximum flow into and minimum flow out of E	AO2.4	E1	$AD = 4$ from (a)(i), so $DE = 1$ and $DT = 3$. Since flow out of vertex E is at least 2, and $DE = 1$, BE must be at its upper capacity of 1.
(b)	Explains correctly the statement using a cut in the network	AO2.4	E1	Min flow across the cut $\{S, A, B, C\} \setminus \{D, E, G, T\}$ $= AD + BE + \min(BG) + \min(CG)$ $= 4 + 1 + 5 + 2$ $= 12 > 11$

(c)	Finds a potential increase and decrease for each arc with a value on the two arrows for each arc, with SA, SC, AB, BC and CF correct.	AO1.1a	M1	See diagram below
(i)	Determines correctly the potential increase and decrease for each arc with values on all arrows correct.	AO1.1b	A1	



(ii)	Finds correctly one augmenting path and flow	AO1.1b	B1	<table><tr><th>Augmenting Path</th><th>Flow</th></tr><tr><td>SADEFT</td><td>1</td></tr><tr><td>SCBET</td><td>3</td></tr><tr><td>SADBET</td><td>1</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	Augmenting Path	Flow	SADEFT	1	SCBET	3	SADBET	1				
	Augmenting Path	Flow														
	SADEFT	1														
SCBET	3															
SADBET	1															
	Finds correctly two augmenting paths and the flows	AO1.1b	B1													
	Finds correctly three (or four) augmenting paths and the flows, and clearly states the maximum flow in context with units	AO3.2a	B1	Maximum flow through the network is 22 litres per second												
(iii)	Constructs a rigorous mathematical proof using the value of a cut and the maximum flow-minimum cut theorem (to achieve this mark, the student must clearly show the cut they are calculating the value of and clearly state that the value of this cut is equal to the value of the flow in (c)(ii) and then conclude that the flow is maximal by the maximum flow-minimum cut theorem)	AO2.1	R1	$\{S, A, B, C, D, E\} / \{F, T\} = 7 + 3 + 9 + 3 = 22$ As the flow (22) is equal to the value of the cut (22), the maximum flow is 22 by the maximum flow-minimum cut theorem.												

(iv)	Argues that, as 7 litres per second are initially flowing through node E, only a further 2 litres per second can flow through node E	AO2.4	R1	Flow through network can only increase by 2 litres per second as DT and CF are already saturated Therefore restricted capacity node reduces the maximum flow to $17 + 2 = 19$ litres per second
------	--	-------	----	--



Interprets the impact of the restricted capacity node, concluding that the maximum flow is reduced to 19 litres per second	AO3.A	R1	
Total 11 marks			

Q2.

- (a) Cut value = $40 + 27 + 0 + 24$
= 91

B1

1

- (b) $ABDX$ 16

B1

$GFBX$ 18

B1

$GHEX$ 20

B1

3

- (c) (i) One correct route with additional flow
any feasible route and flow

M1

Another 2 routes and flows correct
total flow at least 80

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A1

All routes correct with total flow = 85

A1cso

Forward and backward flows on diagram
(directions must be clear)
at least 8 edges with pairs of values 'correct'

M1

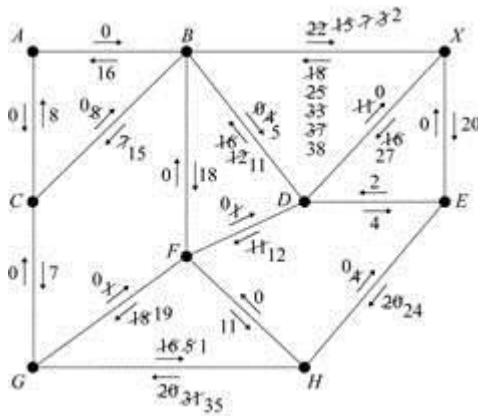
Augmenting flows
correct

A1cso

5

Consider other possible correct flows

Condense diagram as shown but really should have initial flows in DE, etc



Route	Flow
ABDX	16
GFBX	18
GHEX	20
GCBX	7
ACBX	8
GHFDX	11
GHEDBX	4
GFDBX	1

(ii) Max flow = 85

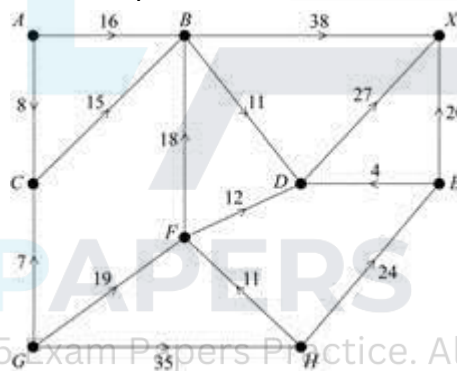
B1

Correct max flow

B1

2

Consider other possible correct flows



(d) Considering 'their' $AB+CB+FB - 45 = 4$ fewer

M1

$\Rightarrow$ max number = 81

A1cao

2

[13]

Q3.

(a) Arrival gates are U and R

B1

1

(b) Cut value = $45 + 53 + 20 + 37 + 0$
= 155

B1

1

(c) Max flow along *UTSP* is 17

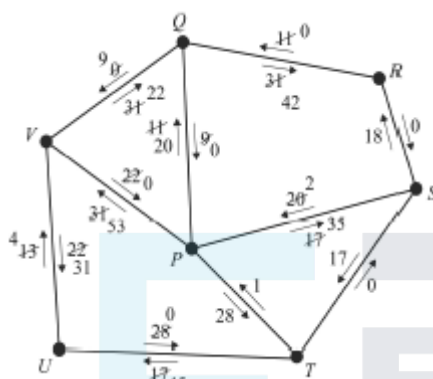
B1

and along *RQVP* is 31

B1

2

(d) (i)



Route	Value of Flow
<i>UTSP</i>	17
<i>RQVP</i>	31
<i>RSP</i>	18
<i>RQP</i>	11
<i>UTP</i>	28
<i>UVP</i>	22
<i>UVQP</i>	9

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Initial flows along *UTSP* and *RQVP* with potential increases and decreases

B1

Table: first route and correct flow

After UTSP and RQVP

M1

Another route and flow

A1

Table correct

A1

Network: attempt to use labelling procedure with forward/backward flows

M1

All diagram correct

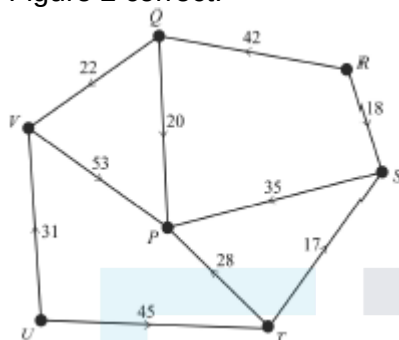
A1

6

(ii) Maximum flow = 136

B1

Figure 2 correct:



Other possible answers

B1

2

(e) Rate reduced by 3
"their" maximum flow – 3

M1

New maximum is 133

A1

2

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[14]

Q4.

(a) $C_1 = 100 + 80 + 150$
 $= 330$

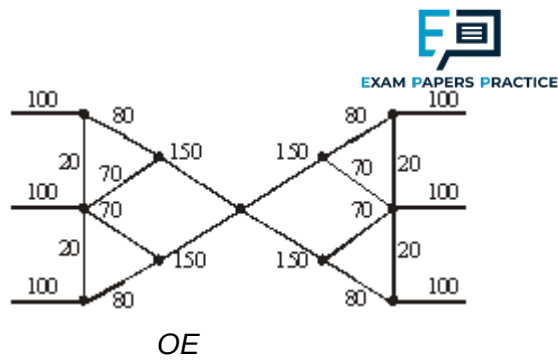
B1

$C_2 = 80 + 80 + 100 + 80 + 80$
 $= 420$

B1

2

(b)



M1A1A1

3

- (c) (i) At I max = 300
 $\therefore$ restriction $\Rightarrow$ max = 120

B1

1

- (ii) At H max = 150

$\therefore$ restriction $\Rightarrow$ max = 270

M1

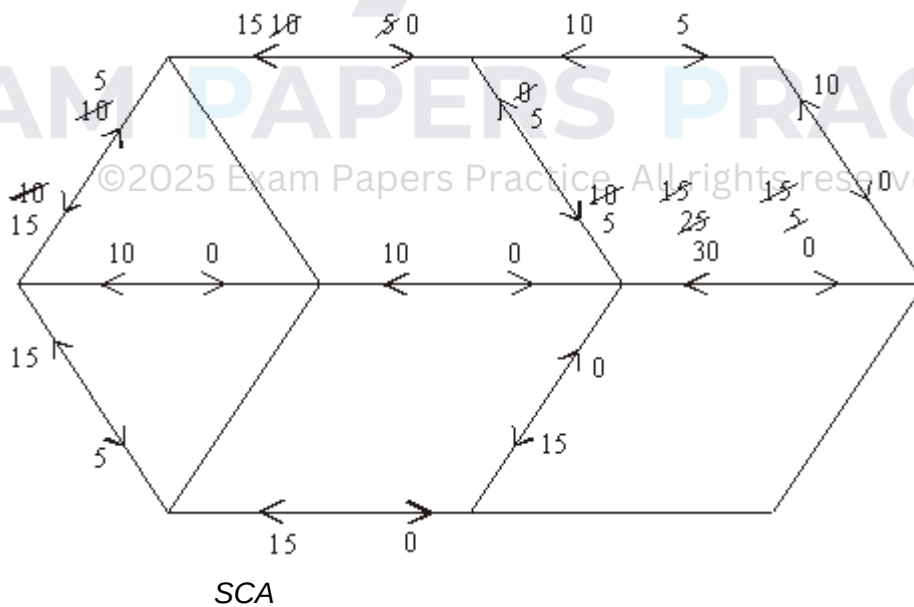
A1

2

[8]

Q5.

(a)



first flow

M1

second flow

A1

A1

$\therefore$ maximum flow = 40

B1

4

(b) Cut BE, DG, CF

B1

1

(c) Out of A – max 50

B1

$\therefore$ from A must be road

Same as J

B1

$\therefore$ join AJ direct

2

[7]

Q6.

(a) A, B, C, D

B1

1

(b) /

B1

1

(c) $20 + 20 + 10 + 10 + 40 + 50 + 30$

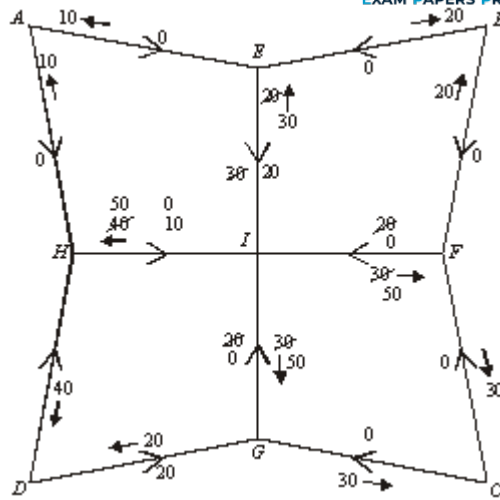
M1

= 180

A1

2

(d)



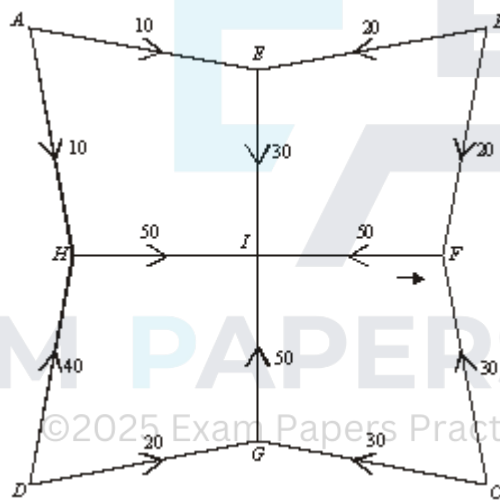
SCA

M1

1 complete flow

M1

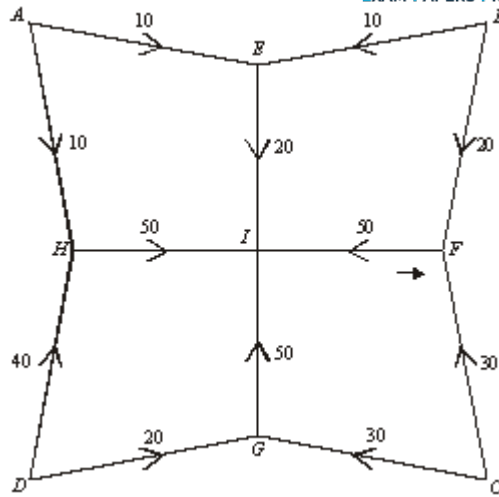
Final:



OE

A1

(e)



Their max – 10

M1

Max flow = 170

CAO

A1

2

[11]

Q7.

(a) $C_1 = 40$

B1

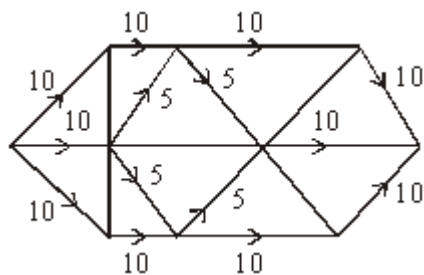
$C_2 = 35$

B1

2

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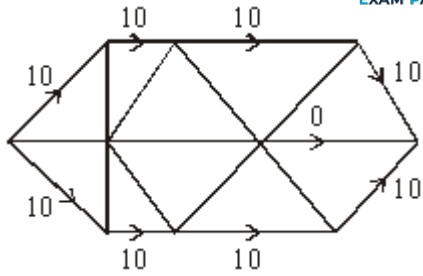
(b)



M1 A1

2

(c)



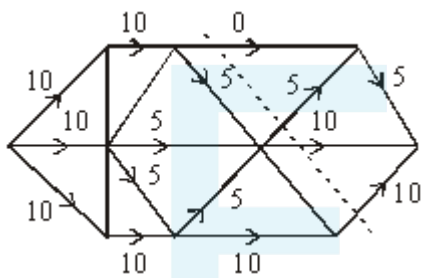
M1

Max flow = 20

A1

2

(d) (i)



M1

Max flow = 25

A1

2

(ii) Min cut = 25
Attempted cut

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M1

Min cut = Max flow

Correct cut GH, GJ, IJ or $(C_2 - EH)$

A1

2

[10]

Q8.

(a) **SC, SB, AT**

M1 A1

2

(b) (i) **SC T 1**

M1 A1

2

M1 A1
2

- (c) *any flow $\leq$ any cut
so any flow ≤ 16*

B1
1

- (d) *Choose AT.*

B1

The arc must be in the minimum cut in

M1

*(a). Also it must be in {AT, BT, CT}
which is a cut of 17.*

A1
3

[10]

Q9.

- (a) e.g. 300

M1

SABT 100, SADCBT 50

A1 A1

SADCEBT 100, SADCBT 50

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A1 A1

Flow of 650

Max of 3 for final flow only

B1
6

- (b) *DE, CE, CB, AB*

M1 A1
2

- (c) *Can augment SDCT by 50*

B1

New network still has a cut of 700 (AB, AC, DC, DE)



EXAM PAPERS PRACTICE

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Examiner reports

Q2.

- (a) Many candidates failed to obtain the correct value of the cut, usually as a result of subtracting 6 instead of adding 0.
- (b) Again, few had trouble finding the maximum flows along the given edges.
- (c) It was good to see candidates trying to set out their solution in a logical manner and once again the diagram and table clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made. Marks are awarded for the initial flow and it is very difficult to credit candidates for their original values if they have been obliterated during augmentation. This was a good discriminator and tested that candidates understood the labelling procedure. Only the best candidates realised the need to augment a flow in the direction from *D* to *B*; not many obtained a flow greater than 80 and very few achieved the correct flow of 85 with a fully augmented correct diagram.
- (d) Most candidates realised that they needed to reduce their total flow by 4 because of the vertex restriction, but only the best candidates obtained the correct maximum number that could move through the network during the fire drill.

Q3.

In part (a) most candidates identified the arrival gates and scored an easy mark.

In part (b) quite a few candidates made errors in calculating the value of the cut, with common wrong answers being 138 and 172.

In part (c) the values of the initial maximum flows were usually correct.

In part (d) it was good to see candidates trying to set out their solution in a logical manner and once again the insert clearly helped. Only a few candidates failed to show potential forward **and** backward flows on their network. Candidates are advised to use the table to show what new flows have been introduced and to modify both the forward and backward flows in their network. The previous values should be clear to the examiner when such modification is made.

For part (d)(ii) the incorrect answer of 127 was seen far more often than the correct value for the maximum flow, since candidates seemed unaware of the need to introduce an additional flow of 9 by actually reducing the flow along some edges that were previously at full capacity.

In part (e) those who had the correct maximum flow in part (d) were usually successful in finding a new maximum flow and those who did not have a correct answer were given some credit for reducing their previous flow by 3 or explaining how the extra 3 passengers could be accommodated in some way.

Q4.

This question proved to be a good discriminator with the best candidates scoring